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GLOW: Capital-Efficiency Positioning

1 A Design Capability Built for the Real Decisions

GLOW is Cartheur's in-house design environment for custom silicon. This paper describes the value that environment can bring to a customer engagement: clearer decisions, faster learning, durable design knowledge, and disciplined investment. It also makes a simple commitment to accuracy: documented facts are presented as facts, while cost, schedule, and performance outcomes are supported by evidence from the relevant program.

2 What GLOW Brings to an Engagement

GLOW is an in-house design environment. Its value is not simply direct access to a CAD tool. It is the design control, responsive iteration, retained knowledge, and thoughtful management of custom-silicon risk that Cartheur brings by owning and evolving its environment.

This distinction matters because the two decisions are different. A commercial CAD buyer evaluates feature coverage, interoperability, support contracts, and licensing. A custom-silicon customer needs to know whether the design team can understand the problem, make sound trade-offs, respond when requirements change, and deliver a manufacturable result with an appropriate non-recurring engineering (NRE) commitment. GLOW supports that second decision.

3 What Capital Efficiency Means

Capital efficiency is not simply lower software expense. It is the ability to turn a limited engineering and financial budget into more validated design options, more reusable know-how, and a shorter path from a customer problem to a manufacturable implementation.

3.1 Design control

Owning the environment gives the engineering team direct control over the tools that express and inspect a design. The team can adapt its workflow to a new architecture, process constraint, or verification question without first waiting for a vendor roadmap. This is valuable when the design problem is unusual or when a late discovery would otherwise require a costly workflow change.

3.2 Faster learning loops

GLOW shortens the path between an architectural decision and the evidence needed to test it. The team can make and evaluate changes within the same design culture, preserving context rather than repeatedly translating the problem between disconnected tools and organizations.

3.3 Reusable knowledge

Methods, checks, libraries, and design conventions compound in value when they remain with the organization. The useful customer outcome is continuity: lessons from one program can inform the next, and the team can retain the reasoning behind a trade-off rather than only its final artifact.

3.4 NRE discipline

The strongest capital-efficiency proposition is disciplined NRE. GLOW directs engineering effort toward the silicon, validation, and customer use case instead of avoidable tool friction. The result is a program built to turn investment into validated design progress.

4 The Customer Value Proposition

For customers considering a custom-silicon program, the value proposition is:

Cartheur uses GLOW as part of its in-house design capability to keep critical design knowledge and iteration authority close to the engineering team. For customers, the benefit is a more adaptable custom-silicon engagement: clearer trade-offs, fewer avoidable hand-offs, and a design partner able to respond directly to the problem.

GLOW gives customers a design capability that stays close to the product question. By keeping architecture, implementation, and engineering judgment connected, Cartheur can move from an ambitious system brief to a manufacturable design without surrendering the context that makes the product distinctive.

4.1 How an Engagement Works

An engagement begins with the system problem, not a tool selection. Cartheur and the customer define the workload, energy envelope, physical constraints, interfaces, manufacturing assumptions, and the decision that makes the program worthwhile. That shared brief becomes the basis for an architectural exploration rather than a hand-off to an isolated implementation team.

GLOW keeps the exploration close to the engineers responsible for the design. When a constraint changes or a new question emerges, the team can revisit the relevant trade-off with the customer context still intact. The goal is not to promise that every change is inexpensive; it is to make the cost, consequence, and available alternatives visible early enough to choose deliberately.

The customer receives a more legible development relationship: a defined problem, explicit trade-offs, reviewable artifacts, and a design partner whose methods carry from concept through implementation. As evidence becomes available, the program can convert those artifacts into reusable design knowledge for a later product or a related application.

4.2 Where the Proposition Is Strongest

The proposition is strongest when a program needs custom silicon but does not fit a standard, data-center-first template: an energy-constrained edge system, a tightly coupled hardware/software design, an unusual I/O or packaging requirement, or a product that benefits from retaining architectural control over time. In those situations, the customer is choosing more than a tool; it is choosing a design capability and a team that can carry that capability forward.

5 The GLOW Silicon Design Flow: *From System Idea to Silicon*

GLOW connects the critical stages of a custom-silicon program in one continuous engineering pipeline. Each stage builds evidence for the next, so that a product team can move forward with greater clarity, control, and confidence.

Stage	GLOW focus	Program outcome
Frame	Define the workload, energy envelope, interfaces, and constraints that make the program worthwhile.	A shared design brief and clear investment decision.
Explore	Test architectural options against the system question while trade-offs are still visible.	Validated options for the product architecture.
Implement	Carry the chosen architecture into a focused custom-silicon design flow.	A direct route from system intent to engineering execution.
Verify	Inspect behavior, assumptions, and design decisions before fabrication.	Earlier confidence in the design and its critical choices.
Prepare	Assemble the artifacts, reviews, and manufacturing assumptions needed for the next engineering gate.	A disciplined path toward manufacturable silicon.
Carry forward	Retain methods, checks, libraries, and the reasoning behind key decisions.	Reusable knowledge for the next product and the next generation.

6 Evidence That Builds Confidence

Custom-silicon decisions should be supported by engineering evidence that can be inspected. Cartheur will demonstrate GLOW through concise, verifiable examples that connect a system challenge to a design decision, the resulting artifact, and a measurable outcome. Depending on what can be shared, that evidence may include an avoided respin, a shorter design iteration, a reusable interface or library, a process migration, or a verification issue found before fabrication.

Customer confidentiality does not have to prevent a meaningful account of the work. Where details cannot be published, Cartheur will identify the process generation, the kind of change, the stage at which it was discovered, and an approved range or percentage for the measured effect. This gives technical readers a basis for assessment while protecting the program behind the result.

7 A Stronger Alternative for Custom Silicon

GLOW is built for custom-silicon programs where architectural control, rapid learning, and close engineering collaboration are decisive. It unites the design environment with the people responsible for the silicon, enabling Cartheur to explore unconventional architectures, respond to changing constraints, and retain the reasoning behind each important trade-off.

GLOW competes where a standard, vendor-centered flow creates distance between the architecture and the team that must make it real. For energy-constrained edge systems, tightly coupled

hardware/software products, unusual I/O or packaging requirements, and designs that demand lasting architectural ownership, it is a focused alternative shaped around the product rather than a generic workflow.

The question for a technical buyer is whether its design partner has the control, technical depth, and responsiveness to turn a difficult product question into manufacturable silicon. GLOW is Cartheur's answer.

8 A Proven Design Heritage

GLOW was used in the development of the GA4 and GA144 multicomputer chips that Volatco ships today. That history matters because it connects the environment to practical silicon work: the design choices, verification discipline, and engineering knowledge behind the tools were developed alongside real multicomputer products, not in isolation.

The GA144 architecture brings together 144 complete computers with integrated memory and I/O, operating without a global clock. GLOW's full-chip simulator and performance charting gave the engineering team a practical way to visualize workload placement, inter-node behavior, and resource use across the array before validation on silicon. These tools supported the exploration of deeply parallel, asynchronous computing—the kind of system-level design work that is difficult to understand one node at a time.

9 Looking Ahead: AI in Your Pocket

Since acquiring the full GreenArrays IP stack, Cartheur Research has been bringing GLOW's power to a wider technical audience. The objective is to help engineers design the next generation of AI chips for capable, energy-conscious systems that live with their users—not only in large data centers, but as **AI in your pocket**.

This work progressively exposes the parts of the design environment that help a technical audience understand, explore, and build practical AI hardware. Each public capability will identify its intended user, supported design stage, and support model, giving engineers a clear basis for adoption.

10 A Path for Public Collaboration

Cartheur's approach is to make GLOW understandable before making it broadly accessible. Technical readers can engage with the design ideas, constraints, and representative examples behind a low-energy AI hardware flow, and see why an approach for compact, energy-conscious systems differs from one optimized primarily for large data centers.

As capabilities are released, researchers and experienced practitioners will be able to explore selected concepts within clear boundaries. Documentation will distinguish experimental material from supported workflows and internal capabilities. Released workflows will address practical questions—workload mapping, architectural trade-offs, and the information needed for an engineering review—with the supported design stage, expected result, and limitations made explicit.

11 Start With the Design Question

The most productive first conversation is about the system a team needs to create, rather than the tools it expects to use. Cartheur welcomes engineers, researchers, and product leaders working on AI hardware where energy use, physical footprint, response time, or long-term architectural control matters.

The starting points are clear: the workload, energy and latency envelope, interfaces, package and process assumptions, and the evidence needed for the next investment decision. These constraints turn an ambitious idea into a design brief. Cartheur's engineers can then frame the architectural options, their trade-offs, and a technically credible route to manufacturable implementation.

The measure of success is not maximum tool exposure. It is a growing technical community that can use released capabilities responsibly to design energy-conscious AI systems that belong close to the people who depend on them.

12 The EDA Landscape

The established EDA market is led by **Synopsys Design Compiler and VCS** for synthesis and simulation, **Cadence Virtuoso and Innovus** for custom design and digital implementation, and **Siemens Calibre** for physical verification. Synopsys, Cadence, and Siemens EDA have built the dominant platforms behind much of today's semiconductor industry.

GLOW competes with a distinct proposition: an integrated design environment for engineers who value architectural control, fast learning loops, and energy-conscious custom silicon. It gives teams with demanding, non-standard hardware goals a focused alternative to the established market—one that keeps the product architecture and the engineering team in direct command of the path to silicon.



Choose GLOW as your next semiconductor design tool.